

Statistical Optimization: Lecture 10

Newton's method

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Outline

Newton's method

1-dimensional Newton method

Newton's method was originally introduced for solving

$$f(\theta) = 0.$$

Starting from $\theta_0 \in \mathbb{R}$, it computes

$$\theta_{t+1} = \theta_t - \frac{f(\theta_t)}{f'(\theta_t)}, \quad t \geq 0.$$

Geometric view: θ_{t+1} is the intersection of the tangent line at $(\theta_t, f(\theta_t))$ with the horizontal axis, i.e. it solves

$$f(\theta_t) + f'(\theta_t)(\theta - \theta_t) = 0.$$

Remark. We can apply this idea to find zero gradient.

1-dimensional Newton method

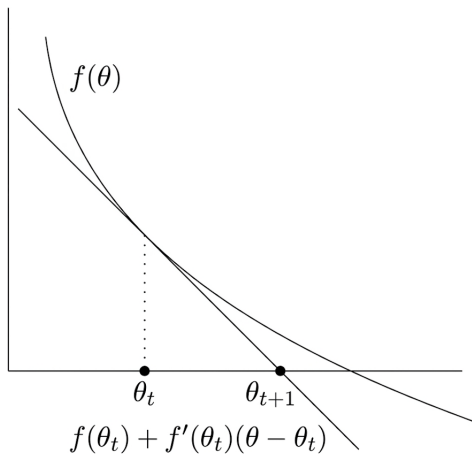


Figure: One step of Newton's method

Newton's method for optimization

Suppose we want to minimize a twice differentiable function

$$f : \mathbb{R} \rightarrow \mathbb{R}.$$

In one dimension, this can be reduced to finding a zero of f' . Applying Newton's method to f' , we obtain

$$\theta_{t+1} := \theta_t - \frac{f'(\theta_t)}{f''(\theta_t)} = \theta_t - (f''(\theta_t))^{-1} f'(\theta_t), \quad t \geq 0. \quad (1)$$

There is no reason to restrict to $d = 1$. For

$$f : \mathbb{R}^d \rightarrow \mathbb{R},$$

Newton's method iterates

$$\theta_{t+1} := \theta_t - (\nabla^2 f(\theta_t))^{-1} \nabla f(\theta_t), \quad t \geq 0. \quad (2)$$

Second-order intuition and gradient descent

To minimize a smooth function f , we approximate it locally near θ_t by a **second-order Taylor expansion**:

$$f(\theta_t + p_t) \approx f(\theta_t) + \nabla f(\theta_t)^\top p_t + \frac{1}{2} p_t^\top \nabla^2 f(\theta_t) p_t.$$

Minimizing this quadratic model (when $\nabla^2 f(\theta_t)$ is positive definite) gives

$$p_t = -(\nabla^2 f(\theta_t))^{-1} \nabla f(\theta_t).$$

If we replace the Hessian by a scaled identity matrix,

$$\nabla^2 f(\theta_t) \approx \frac{1}{\gamma_t} I,$$

then minimizing this simplified model yields

$$p_t = -\gamma_t \nabla f(\theta_t).$$

which is exactly the gradient descent update.

First-order and second-order intuition

To minimize a smooth function f , we approximate it locally near θ_t by **Taylor expansion**.

First-order approximation:

$$f(\theta_t + p_t) \approx f(\theta_t) + \nabla f(\theta_t)^\top p_t.$$

This uses only gradient information, so a natural direction is

$$p_t = -\nabla f(\theta_t).$$

Second-order approximation:

$$f(\theta_t + p_t) \approx f(\theta_t) + \nabla f(\theta_t)^\top p_t + \frac{1}{2} p_t^\top \nabla^2 f(\theta_t) p_t.$$

This also uses curvature information. With some conditions we minimize this quadratic model suggests

$$p_t = -(\nabla^2 f(\theta_t))^{-1} \nabla f(\theta_t).$$

Newton step minimizes the quadratic model

Theorem

Let f be convex and twice differentiable at θ_t , and assume

$$\nabla^2 f(\theta_t) \succ 0$$

is invertible. Then the Newton iterate θ_{t+1} satisfies

$$\theta_{t+1} = \arg \min_{\theta \in \mathbb{R}^d} \left[f(\theta_t) + \nabla f(\theta_t)^\top (\theta - \theta_t) + \frac{1}{2} (\theta - \theta_t)^\top \nabla^2 f(\theta_t) (\theta - \theta_t) \right].$$

Proof of the theorem

Proof. Consider the quadratic function

$$m_t(\theta) := f(\theta_t) + \nabla f(\theta_t)^\top (\theta - \theta_t) + \frac{1}{2}(\theta - \theta_t)^\top \nabla^2 f(\theta_t)(\theta - \theta_t).$$

Since $\nabla^2 f(\theta_t) \succ 0$, the function m_t is strictly convex and has a unique minimizer. Setting its gradient to zero gives

$$\nabla m_t(\theta) = \nabla f(\theta_t) + \nabla^2 f(\theta_t)(\theta - \theta_t) = 0.$$

Hence

$$\theta = \theta_t - (\nabla^2 f(\theta_t))^{-1} \nabla f(\theta_t) = \theta_{t+1}.$$

Therefore θ_{t+1} is the unique minimizer of m_t .



Theorem (Quadratic Convergence)

Let $f : \text{dom}(f) \rightarrow \mathbb{R}$ be twice differentiable with a critical point θ^* . Suppose there is a ball $\Theta \subseteq \text{dom}(f)$ centered at θ^* such that:

(i) Bounded inverse Hessians: $\exists \mu > 0$ with

$$\|(\nabla^2 f(\theta))^{-1}\| \leq \frac{1}{\mu}, \quad \forall \theta \in \Theta.$$

(ii) Lipschitz continuous Hessians: $\exists B \geq 0$ with

$$\|\nabla^2 f(\theta) - \nabla^2 f(\eta)\| \leq B\|\theta - \eta\|, \quad \forall \theta, \eta \in \Theta.$$

(Here $\|\cdot\|$ is the spectral norm.) Then for $\theta_t \in \Theta$ and θ_{t+1} given by Newton's method,

$$\|\theta_{t+1} - \theta^*\| \leq \frac{B}{2\mu} \|\theta_t - \theta^*\|^2.$$

Proof. By Newton's method,

$$\theta_{t+1} - \theta^* = \theta_t - \theta^* - (\nabla^2 f(\theta_t))^{-1} \nabla f(\theta_t).$$

Since $\nabla f(\theta^*) = 0$, we expand $\nabla f(\theta^*)$ around θ_t :

$$\nabla f(\theta^*) = \nabla f(\theta_t) + \int_0^1 \nabla^2 f(\theta_t + s(\theta^* - \theta_t)) (\theta^* - \theta_t) ds.$$

Therefore,

$$0 = \nabla f(\theta_t) + \nabla^2 f(\theta_t) (\theta^* - \theta_t) + \int_0^1 \left(\nabla^2 f(\theta_t + s(\theta^* - \theta_t)) - \nabla^2 f(\theta_t) \right) (\theta^* - \theta_t) ds.$$

Rearranging gives

$$\nabla f(\theta_t) = \nabla^2 f(\theta_t) (\theta_t - \theta^*) - \int_0^1 \left(\nabla^2 f(\theta_t + s(\theta^* - \theta_t)) - \nabla^2 f(\theta_t) \right) (\theta^* - \theta_t) ds.$$

Substituting this identity into the Newton step, we obtain

$$\theta_{t+1} - \theta^* = (\nabla^2 f(\theta_t))^{-1} \int_0^1 \left(\nabla^2 f(\theta_t + s(\theta^* - \theta_t)) - \nabla^2 f(\theta_t) \right) (\theta^* - \theta_t) ds.$$

Taking norms, and using the spectral norm, gives

$$\|\theta_{t+1} - \theta^*\| \leq \|(\nabla^2 f(\theta_t))^{-1}\| \int_0^1 \left\| \left(\nabla^2 f(\theta_t + s(\theta^* - \theta_t)) - \nabla^2 f(\theta_t) \right) (\theta^* - \theta_t) \right\| ds.$$

Hence

$$\|\theta_{t+1} - \theta^*\| \leq \|(\nabla^2 f(\theta_t))^{-1}\| \|\theta^* - \theta_t\| \int_0^1 \left\| \nabla^2 f(\theta_t + s(\theta^* - \theta_t)) - \nabla^2 f(\theta_t) \right\| ds.$$

By assumptions (i) and (ii),

$$\|\theta_{t+1} - \theta^*\| \leq \frac{1}{\mu} \|\theta^* - \theta_t\| \int_0^1 B \|s(\theta^* - \theta_t)\| ds \leq \frac{B}{2\mu} \|\theta_t - \theta^*\|^2.$$

Therefore,

$$\|\theta_{t+1} - \theta^*\| \leq \frac{B}{\mu} \|\theta^* - \theta_t\|^2 \int_0^1 s ds = \frac{B}{2\mu} \|\theta_t - \theta^*\|^2.$$

Convergence of Newton's method

Corollary (Convergence of Newton's method)

With the assumptions of Theorem, if $\theta_0 \in \Theta$ satisfies

$$\|\theta_0 - \theta^*\| \leq \frac{\mu}{B},$$

then Newton's method yields

$$\|\theta_T - \theta^*\| \leq \frac{\mu}{B} \left(\frac{1}{2}\right)^{2^T - 1}, \quad T \geq 0.$$

Proof. Let $\mathbf{e}_t := \|\theta_t - \theta^*\|$. Theorem gives $\mathbf{e}_{t+1} \leq \frac{B}{2\mu} \mathbf{e}_t^2$. Assuming $\mathbf{e}_t \leq \frac{\mu}{B} \left(\frac{1}{2}\right)^{2^t - 1}$,

$$\mathbf{e}_{t+1} \leq \frac{B}{2\mu} \left(\frac{\mu}{B}\right)^2 \left(\frac{1}{2}\right)^{2(2^t - 1)} = \frac{\mu}{B} \left(\frac{1}{2}\right)^{2^{t+1} - 1}.$$